The documentation and process conversion measures necessary to comply with this document shall be completed by 15 May 2019.

INCH-POUND

MIL-PRF-19500/646F
15 February 2019
SUPERSEDING
MIL-PRF-19500/646E
26 November 2008

PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, ULTRAFAST, TYPES 1N6774 THROUGH 1N6777, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

* 1.1 This specification covers the performance requirements for silicon, ultrafast, power rectifier diode. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500.

* 1.2 Package outlines and die topography. The device package for the encapsulated device type are as follows: (2 pin, isolated - TO-257) in accordance with figure 1.

1.3 Maximum ratings. $T_C$ = 25°C unless otherwise specified.

| Types | $V_{RWM}$ (1) $I_D$ = 10 µA dc | $I_F$ (1) (2) $T_C$ =+100°C | $I_{FSM}$ (1) $t_p$= 8.3 ms | $R_{\theta JC}$ (1) | $R_{\theta JA}$ (1) | $T_{STG}$ and $T_J$ |
|---|---|---|---|---|---|---|
| | Vdc | A dc | A (pk) | °C/W | °C/W | °C |
| 1N6774 | 50 | | | | | |
| 1N6775 | 100 | 15 | 180 | 2.0 | 40 | -65 to +150 |
| 1N6776 | 150 | | | | | |
| 1N6777 | 200 | | | | | |

(1) Each individual diode.
(2) Derate at 300 mA/°C above $T_C$ = +100°C.

1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics are at +25°C, and for each diode.

| Types | $V_{F1}$ $I_F$ = 8 A dc | $V_{F2}$ $I_F$ =15 A dc | $I_{R1}$ (see 1.3) $V_R$ = 0.8 $V_{RWM}$ | $I_{R2}$ $V_R$ = 0.8 $V_{RWM}$ (see 1.3) $T_C$ = +100°C | $t_{rr}$ | $C_J$ $V_R$ = 5 V f = 1 MHz |
|---|---|---|---|---|---|---|
| | V dc | V dc | µA dc | µA dc | ns | pF |
| All devices | 1.00 | 1.15 | 10 | 500 | 35 | 300 |

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

* 1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

* 1.5.2 Device type. The designation system for the device types of semiconductors covered by this specification sheet are as follows.

* 1.5.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

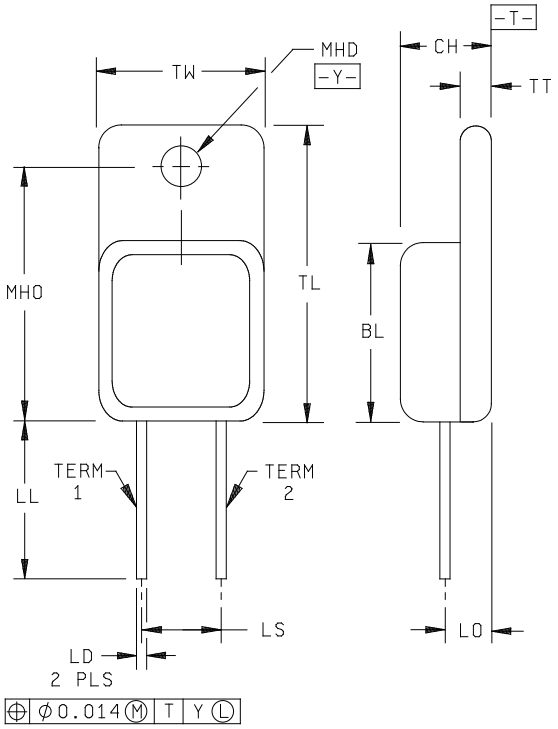
* 1.5.2.2 Second number symbols. The second number symbols for the semiconductors covered by this specification sheet are as follows: "6774", “6775”, “6776”, and "6777".

* 1.5.3 Suffix symbols. Suffix symbols are not applicable for this specification sheet.

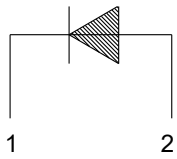
* 1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.



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SCHEMATIC




Configuration

| Terminal | Description |
|---|---|
| 1 | Cathode |
| 2 | Anode |

| Dimensions | | | | |
|---|---|---|---|---|
| Symbol | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .410 | .430 | 10.4 | 10.9 |
| CH | .249 | .260 | 6.32 | 6.60 |
| LD | .035 | .045 | 0.89 | 1.14 |
| LL | .500 | .750 | 12.70 | 19.05 |
| LO | .150 typ | | 3.81 typ | |
| LS | .200 bsc | | 5.08 bsc | |
| MHD | .140 | .150 | 3.55 | 3.80 |
| MHO | .527 | .537 | 13.4 | 13.6 |
| TL | .645 | .665 | 16.4 | 16.9 |
| TT | .040 | .050 | 1.02 | 1.27 |
| TW | .410 | .420 | 10.4 | 10.7 |

FIGURE 1. Physical dimensions and configuration (2 pin, isolated) (TO-257).



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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://quicksearch.dla.mil.

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be group A as specified herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.



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3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANTX, JANTXV, and JANS levels). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (1) 3c | Thermal impedance (see 4.3.2) | Thermal impedance (see 4.3.2) |
| 9 and 10 | Not applicable | Not applicable |
| 11 | $I_{R1}$ and $V_{F1}$ | $I_{R1}$ and $V_{F1}$ |
| 12 | See 4.3.1, t = 240 hours | See 4.3.1, t = 48 hours |
| 13 | Subgroups 2 and 3 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial value or ±2.5 μA, whichever is greater; $\Delta V_{F1} \leq$ ±100 mV. | Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial value or ±2.5 μA whichever is greater; $\Delta V_{F1} \leq$ ±100 mV. |

* (1) Thermal impedance shall be performed any time after temperature cycling, screen 3a, and the JANTX and JANTXV do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038 of MIL-STD-750, test condition A. $T_J$ = +125°C; $V_R$ = 0.8 of rated $V_{RWM}$ (see 1.3).

4.3.2 Thermal impedance The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD-750, using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, ($V_C$ and $V_H$ where appropriate). See table II, subgroup 4 herein.



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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I, herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIa (JANS) and table E-VIb (JANTX and JANTXV) of MIL-PRF-19500 and as follows. Delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, appendix E, table E-VIa (JANS) of MIL-PRF-19500.

| | Subgroup | Method | Condition |
|---|---|---|---|
| * | B4 | 1037 | $I_F$ or $I_O$ = 1.25 A to 10 A; $\Delta T_J$ = +85°C minimum. |

* 4.4.2.2 Group B inspection, appendix E, table E-VIb (JANTX and JANTXV) of MIL-PRF-19500.

| | Subgroup | Method | Condition |
|---|---|---|---|
| * | B3 | 1037 | $I_F$ or $I_O$ = 1.25 A to 10 A; $\Delta T_J$ = +85°C minimum. |
| * | B5 | 4081 | $R_{\theta JC}$, See 1.3 and 4.5.2. |

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500 and as follows. Delta requirements shall be in accordance with the applicable steps of table III herein.

* 4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

| | Subgroup | Method | Condition |
|---|---|---|---|
| | C2 | 2036 | Test condition A, 5 pounds, t = 15 seconds ±3 seconds. |
| * | C5 | 4081 | See 1.3 and 4.5.2, $R_{\theta JC} \leq 2°$ C/W. |
| * | C6 | 1037 | $I_F$ or $I_O$ = 1.25 A to 10 A; $\Delta T_J$ = +85°C minimum. |

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table II herein. Delta requirements shall be in accordance with the applicable steps and footnotes of table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

* 4.5.2 Thermal resistance. Thermal resistance measurement shall be performed in accordance with method 4081 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, and $t_{SW}$. See, table E-IX, group E, Subgroup 4 of MIL-PRF-19500.



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* TABLE I. Group A inspection. 1/ 2/

| | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| | Subgroup 1 | | | | | | |
| | Visual and mechanical examination | 2071 | | | | | |
| | Subgroup 2 | | | | | | |
| * | Thermal impedance 3/ | 3101 | See 4.3.2. | $Z_{\theta JX}$ | | | °C/W |
| * | Breakdown voltage<br>1N6774<br>1N6775<br>1N6776<br>1N6777 | 4022 | $I_R$ = 10 µA dc, pulsed. 4/ | $V_{BR}$ | <br>50<br>100<br>150<br>200 | | V dc |
| * | Forward voltage | 4011 | Condition B, $I_F$ = 8 A dc, pulsed. 4/ | $V_{F1}$ | | 1.00 | V dc |
| * | | | Condition B, $I_F$ = 15 A dc, pulsed. 4/ | $V_{F2}$ | | 1.15 | V dc |
| * | Reverse leakage current | 4016 | DC or equivalent pulsed method; $V_R$ = 0.8 of $V_{RWM}$ (see 1.3). | $I_{R1}$ | | 10 | µA dc |
| | Subgroup 3 | | | | | | |
| | High temperature operation: | | $T_C$ = +100°C minimum. | | | | |
| * | Reverse leakage current | 4016 | DC or equivalent pulsed method; $V_R$ = 0.8 of $V_{RWM}$ (see 1.3). | $I_{R2}$ | | 500 | µA dc |
| | Low temperature operation: | | $T_A$ = -55°C minimum. | | | | |
| * | Forward voltage | 4011 | Condition B, $I_F$ = 15 A dc, pulsed. 4/ | $V_{F3}$ | | 1.3 | V dc |
| | Subgroup 4 | | | | | | |
| * | Scope display evaluation 5/ | 4023 | | | | | |
| | Reverse recovery time measurements | 4031 | Condition B1. $I_F$ = .5A, $I_{RM}$ = 1A, $I_{R\ (REC)}$ = .25A | $t_{rr}$ | | 35 | ns |

See footnotes at end of table.



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* TABLE I. Group A inspection – Continued. 1/ 2/

| Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroups 5 and 6<br><br>Not applicable<br><br>Subgroup 7<br><br>Junction capacitance | 4001 | $V_R$ = 5 V dc; f = 1.0 MHz. | $C_J$ | | 300 | pF |

1/ For sampling plan, see MIL-PRF-19500.
2/ Each individual diode.
* 3/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
* Group B, subgroup 4 (JANS).
Group C, subgroup 2 and 6.
* Group E, subgroup 1
* 4/ Pulse test: Pulse width = 300 μs, duty cycle ≤ 2 percent.
5/ The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 50 to 100 μA/division and 50 to 100 V/division. Reverse current over the knee shall be at least 500 μA. Each device may exhibit a slightly rounded characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.



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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

| Inspection | MIL-STD-750 | | Sampling plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices c = 0 |
| Temperature cycling | 1051 | 500 cycles | |
| Hermetic seal | 1071 | | |
| * Fine leak | | | |
| * Gross leak | | | |
| Electrical measurements | | See table III herein, steps 1, 2, 3, 4, and 5 | |
| Subgroup 2 | | | |
| Steady-state reverse bias | 1038 | Test condition A, t = 1,000 hours, $T_J$ = +125°C, $V_R$ = 0.8 of rated $V_{RWM}$ (see 1.3) | 45 devices c = 0 |
| Electrical measurements | | See table III herein, steps 1 and 2 | |
| Subgroup 4 | | | |
| Thermal impedance curves | | See MIL-PRF-19500 | |
| Subgroup 5 | | | |
| Not applicable | | | |



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TABLE III. Groups A, B, C, and E electrical and delta measurements. 1/ 2/ 3/ 4/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| ~~1~~ | ~~Forward voltage~~ | ~~4011~~ | ~~$I_F$ = 15 A dc pulsed~~ | ~~$\Delta V_{F2}$~~ | | ~~±100 mV dc from initial reading~~ | |
| ~~2~~ | ~~Reverse leakage current~~ | ~~4016~~ | ~~$V_R$ = 0.8 of $V_{RWM}$ (see 1.3), dc method, pulsed~~ | ~~$\Delta I_{R1}$~~ | | ~~100 percent of initial value or ±2.5 µA dc whichever is greater.~~ | |
| 3 5/ | Reverse recovery time | 4031 | See table I, subgroup 4 herein | $t_{rr}$ | | 34 | ns |

1/ The electrical measurements for table E-VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 4, see table III herein, steps 1, 2, and 3.
   b. Subgroup 5, see table III herein, steps 1, 2, and 3.
2/ The electrical measurements for table E-VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, step 3.
   b. Subgroup 3, see table III herein, step 3.
3/ The electrical measurements for table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroup 6, see table III herein, step 3 for all levels.
4/ The electrical measurements for table E-IX of MIL-PRF-19500 are as follows:
   a. Subgroup 1, see table III herein, steps 1, 2, and 3.
5/ Step 6 applies to irradiated devices only.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

* d. The complete Part or Identifying Number (PIN), see 1.5 and 6.6.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 Cross reference list. Parts covered by this specification may be used to replace the following commercial Part or Identifying Numbers (PIN):

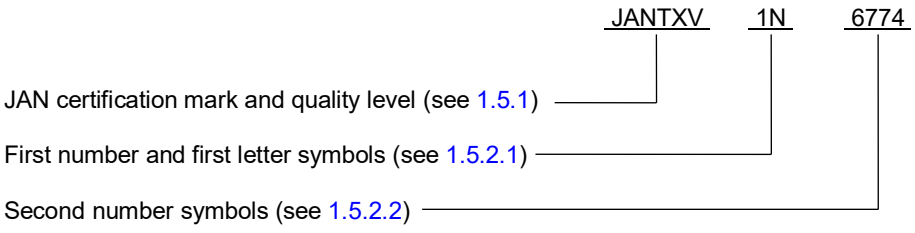
| Preferred types | Commercial types |
|---|---|
| JAN1N6774, JANTX1N6774, JANTXV1N6774,JANS1N6774<br>JAN1N6775, JANTX1N6775, JANTXV1N6775,JANS1N6775<br>JAN1N6776, JANTX1N6776, JANTXV1N6776,JANS1N6776<br>JAN1N6777, JANTX1N6777, JANTXV1N6777,JANS1N6777 | 1N6774<br>1N6775<br>1N6776<br>1N6777 |



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* 6.5 PIN construction example.

* 6.5.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.




* 6.6 List of PINs. The following is a list of possible PINs available on this specification sheet.

| PINs for types 1N6774, 1N6775, 1N6776, AND 1N6777. | | | |
|---|---|---|---|
| JAN1N6774<br>JAN1N6775<br>JAN1N6776<br>JAN1N6777 | JANTX1N6774<br>JANTX1N6775<br>JANTX1N6776<br>JANTX1N6777 | JANTXV1N6774<br>JANTXV1N6775<br>JANTXV1N6776<br>JANTXV1N6777 | JANS1N6774<br>JANS1N6775<br>JANS1N6776<br>JANS1N6777 |

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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